SUPPLEMENTAL MATERIAL

Supplemental Table S1. Search Terms Used

Population	"running" OR "runners"
	AND
Outcome	"injury" OR "injuries" OR "injured"
	AND
Variables	"rearfoot" OR "rear-foot" OR "midfoot" OR "mid-foot" OR "forefoot" OR "fore-foot" OR "foot contact angle" OR "foot angle" OR "foot strike pattern" OR "foot strike angle" OR "foot impact angle" OR "strike index"

Supplemental Table S2. Inclusion and Exclusion Criteria

Inclusion
Studies which have researched running populations (novice, recreational, military and collegiate levels).
Studies which have looked at foot strike pattern at initial contact, foot contact angles or strike index at impact during running trials.
Studies which have explored running-related injuries and how the kinematics of the foot affect this.
Studies which compared injured participants to controls.
Studies written in the English language.
Studies which are fully published papers/journal articles.
Exclusion
Studies which have researched cohorts that are not exclusively runners.
Studies which examined upper limb musculoskeletal running injuries.
Studies that are not written in the English language.
Studies that have been published as conference proceedings or abstracts.
Studies that are opinion articles.

Supplemental Table S3. Risk-of-Bias Assessment*

Criteria	Description
1. Aim clearly described	The aim/hypothesis/objective is clearly described
2. Outcomes described	The main outcomes to be measured are clearly described in the introduction or methods section
3. Subjects clearly described	The characteristics of the subjects included in the trial are clearly described. If running experience was deemed insufficiently described, this was answered no
4. Interventions clearly described	Each intervention to be completed is clearly described
5. Distribution of confounders described	Confounding factors are clearly described. Confounders to be considered include subject's sex, age, weight, running experience, running speed, and foot strike
6. Main findings clearly described	Simple outcome data are reported for all major findings so the reader can check the major analysis and conclusions
7. Estimates of random variability in data	In non-normally distributed data, the interquartile range of results should be reported. In normally distributed data, standard deviations or confidence intervals should be reported
8. All important adverse events reported	The study demonstrates a comprehensive attempt to record all adverse events. This could include discomfort associated with any running condition or delayed onset of muscle soreness
9. Actual probability values reported	Actual probability values (e.g., not $p<0.05$) have been reported for the main outcomes, except where the probability value is less than .001
10. Subjects asked are representative of the population	The source population for subjects and how they were selected are described. Subjects would be representative if they comprised the entire population, an unselected sample of consecutive subjects, or a random sample. Where the study does not report the proportion of the source population from which the subjects are derived, the answer is no
11. Subjects representative of population	The subjects prepared to participate are representative of the entire population from which they were recruited. Validation that the sample was representative would include demonstrating that the distribution of confounding factors was the same in the study sample and the source population

12. Examiners blinded	There was an attempt to blind those measuring the main outcomes
13. Data dredging	Any analysis that had not been planned at the outset of the study is clearly described. If no retrospective unplanned subgroup analysis is reported, the answer is yes
14. Appropriate statistical tests	The statistical tests used to assess the main outcomes are appropriate
15. Valid and reliable main outcome measures	For studies where the outcome measures are clearly described, the answer is yes. For studies that refer to other work or that demonstrate the outcome measures are accurate, the answer is yes
16. Subjects recruited from same population	The subjects in different intervention groups were recruited from the same population. If the subjects acted as their own control, this was answered yes
17. Subjects recruited over same time period	The subjects in different intervention groups were recruited over the same time period. If the subjects acted as their own control, this was answered yes
18. Intervention order randomised	The order of the intervention tested was randomized
19. Adequate adjustment for confounding	There was adequate adjustment for confounding in the analysis from which the main findings were drawn. If the effect of the main confounders was not investigated or confounding was demonstrated but no adjustment was made in the final analysis, this was answered no
20. Sufficient power	If the study reported a power calculation, this was answered yes

*Questions 1-17, 19 and 20 were used to assess risk of bias in the primary review (Relationship between foot strike technique and RRI). Questions 1-20 were used to assess risk of bias in the secondary review (Effect of foot strike technique intervention on risk of RRI); *Adapted from Downs and Black [16].*

Supplemental Table S4. Quality assessment of studies included in the primary review (Foot strike technique and running-related injuries)

Question	*1*	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*10*	*11*	*12*	*13*	*14*	*15*	*16*	*17*	*18*	*19*	*Total*
Retrospective Studies																				
Donoghue et al[15]	1	1	1	1	0	1	1	1	1	0	0	0	1	1	1	0	0	0	0	11
Goss and Gross[23]	1	1	1	1	1	0	1	1	1	0	0	0	1	1	0	1	1	0	0	12
Daoud et al[11]	1	1	1	1	1	1	1	0	1	0	1	0	1	1	1	1	1	1	0	15
Mann et al[45]	1	1	1	1	1	1	1	1	1	0	1	0	1	1	1	1	1	1	1	17
Warr et al[67]	1	1	1	1	0	1	1	1	1	0	1	1	1	1	1	1	1	0	0	15
Paquette et al[53]	1	1	1	1	1	1	1	1	1	0	0	0	1	1	1	0	0	0	1	13
Hollander et al[32]	1	1	1	1	1	1	1	1	1	0	1	0	1	1	1	1	1	0	0	15
Fukusawa et al[21]	1	1	1	1	1	1	1	1	0	0	1	0	1	1	1	1	1	1	1	16
Dingenan et al[14]	1	1	1	1	1	1	1	1	1	0	1	0	1	1	1	0	0	1	0	15
Sugimoto et al[62]	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	0	0	1	0	15
Prospective Studies																				
Kuhman et al[36]	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	0	0	16
Dudley et al[18]	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	0	17
Messier et al[48]	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	0	17

Modified Risk of Bias Assessment Questions: 1:Aim clearly described; 2: Outcomes described; 3: Subjects clearly described; 4: Interventions clearly described; 5: Distribution of confounders clearly described; 6: Main findings clearly described; 7: Estimates of random variability in data; 8: All important adverse events reported; 9: Actual probability values reported; 10: Subjects asked are representative of population; 11: Subjects representative of population; 12: Examiners blinded; 13: Data dredging; 14: Appropriate statistical tests; 15: Valid and reliable main outcome measures; 16: Subjects recruited from same population 17: Subjects recruited over same time period; 18: Adequate adjustment for confounding; 19: Sufficient power.

Supplemental Table S5. Summary of Study Characteristics

Study Characteristics			Participant Characteristics							Testing Characteristics		
Author (Year of Publication)	**Sample size and duration of injury surveillance**	**Definition of injury**	**Running group**	**Age (yrs)**	**Sex (%)**	**BMI (kg/m^2)**	**FSP Prevalence**	**No. included/ analysed (% included)**	**No. injured : No. uninjured**	**Surface**	**Speed**	**Foot strike classification**
Prospective												
Dudley et al[18] (2017)	n = 31. 1 collegiate cross-country season	"Any musculoskeletal complaint of the lower extremities or back causing the restriction of participation in one full practice session".	Collegiate cross-country	20	M: 48% F: 52%	N/R	N/A	32/31 (97%)	12 : 19	Overground in lab	SSS (Mean test speed: 3.9m/s)	FCA: 3D motion capture analysis (5 foot strikes) Not categorized into FSP
Kuhman et al[36] (2016)	n = 19. 1 collegiate cross-country season	N/R	Collegiate cross-country	20 ± 2	M: 58% F: 42%	20.3 ± 1.2	N/A	19/24 (79%)	10 : 9	Overground in lab	4.0-4.5m/s	SI: Force plate (5 foot strikes) Not categorized into FSP
Messier et al[48] (2018)	n = 300. 2 years	"Grade 1: maintaining full activity in spite of symptoms; Grade 2: reducing weekly mileage and; Grade 3: interrupting all training for at least 2 weeks".	Recreational	41 ± 10	M: 57% F: 43%	24.2 ± 3.4	N/A	300/300 (100%)	199 :101	Overground in lab	SSS (Mean test speed: N/R)	SI: Force plate (3 foot strikes) Not categorized into FSP
Retrospective												
Daoud et al (2012)[11]	n = 52. 5 collegiate cross-country seasons	"Full: athlete continues running without restrictions; >50%: athlete runs at a reduced intensity or distance, greater than half of normal training; <50%: athlete runs at a reduced intensity or distance, less than half of normal training; Cross-training: athlete is not running, but is cross-training; Off: athlete is either running nor cross-training".	Collegiate cross-country	19 ± 1	M: 56% F: 44%	20.5 ± 1.5	RFS: 69% MFS: 0% FFS: 31%	52/52 (100%)	N/R	Treadmill and Outdoor track	Treadmill: 3-5m/s, Outdoor: SSS	FSP: Video camera (3 foot strikes) FSP Categorization: Visual^
Goss and Gross (2012)[23]	n = 881. 1 year	"Something that caused you to modify your training schedule for at least 1 week due to pain or discomfort (with or without medical care)".	Recreational, collegiate and military	38 ± 9	M: 50% F: 50%	N/R	RFS: 31% MFS: 43% FFS: 20% Unsure: 6%	904/881 (97%)	881 : 0	N/A	N/A	FSP: Self-reported (foot strikes N/R) FSP Categorization: N/R

Warr et al (2015)[67]	n = 341. 5 years and over a lifetime	"Any injury over the course of the participants life that caused them to modify their training schedule for at least 1 week due to pain or discomfort (with or without formal medical care)".	Military	25 ± 5	M: 100% F: 0%	N/R	RFS: 87% MFS: 9% FFS: 4%	341/341 (100%)	138 : 203	Outdoor runway	Pace of a 2 mile run event (Mean test speed: 3.6m/s)	FSP: video camera (2 foot strikes) FSP Categorization: Visual^
Hollander et al (2021)[32]	n = 550. 7 years	"Any musculoskeletal pain (i.e. muscles, bones, tendons, ligaments) to the lower body that required medical attention". Lower back injuries were also included.	Recreational and Competitive	37 ± 13	M: 50% F: 50%	23.3 ± 3.0	RFS: 71% MFS: 10% FFS: 19%	550/550 (100%)	550 : 0	Treadmill	SSS (Mean test speed: 2.1m/s)	FSP: video camera (10 foot strikes) FSP Categorization: Visual^
Fukusawa et al (2020)[21]	n = 122. N/R	"Presence of diffuse and intermittent pain on the anterior part of the knee for at least 3 months."	Recreational	37 ± 10	M: 70% F:30%	24.6 ± 2.6	RFS: 95% Non-RFS: 5%	122/124 (98%)	60 : 62	Outdoor runway	SSS (Mean test speed: 2.6m/s)	FSP: video camera (5 foot strikes) FSP Categorization: Visual Categories not defined in manuscript text
Sugimoto et al (2019)[62]	n = 70. 5 years	Diagnosis of a hamstring strain which occurred during running, diagnosed by physical examination and MRI.	N/R	29 ± 13	M: 34% F: 66%	21.8 ± 2.6	RFS: 59% MFS: 20% FFS: 51%	71/70 (100%)	35 : 35	Treadmill	PDS between 1.8 – 2.3m/s	FSP: video camera (10 – 60 seconds) FSP Categorization: Visual^
Donoghue et al (2008)[15]	n = 22. 1 year	Presence of chronic Achilles tendon injury	Recreational	42 ± 8	M: 91% F: 9%	N/R	N/R	22/22 (100%)	11 : 11	Treadmill	SSS (Mean test speed: 2.7m/s)	AFA: 3D motion capture analysis (5 foot strikes) FSP Categorization: N/R
Paquette et al (2017)[53]	n = 44. 1 year	N/R	Recreational	29 ± 8	M: 55% F: 45%	22.4 ± 2.4	RFS: 36% Non-RFS: 64%	44/44 (100%)	23 : 21	Treadmill	75% of their 10km PB pace (Mean test speed: RFS 3.4m/s, Non-RFS 2.8m/s)*	FCA: 3D motion capture analysis (5 foot strikes) FSP Categorization: N/R

Dingenen et al (2019)[14]	n = 42 2 years	"Presence of a current running-related knee injury, which caused a restriction of or cessation of running (distance, speed, duration, or training) for at least 7 days or 3 consecutive scheduled training sessions, and that required the runner to consult a physician or other health professional."	Recreational	31 ± 8	M: 29% F: 71%	22.4 ± 2.2	N/R	42/42 (100%)	18 : 24	Treadmill	SSS (Mean test speed: 2.8m/s)	FCA: video camera (7 foot strikes) Not categorized into FSP
Mann et al (2015)[45]	n = 90. 1 year	"A physical pain or complaint altering or interrupting running activity for at least 1 week, affecting the lower extremities and progressive in nature".	Recreational	41 ± 9	M: 73% F: 27%	22.9 ± 2.3	RFS: 93% Non-RFS: 7%	90/90 (100%)	44 : 46	Treadmill	SSS (Mean test speed: 3.0m/s)	SI: Pressure sensitive insoles (161 ± 12 foot strikes) FSP Categorization: N/R

*Values are presented as mean ± standard deviation unless otherwise specified; yrs: years; SSS: self-selected speed; PDS: pre-determined speed; n: number of participants; kg/m^2: kilogram per metre squared; FSP: foot strike pattern; m/s: metres per second; N/R: not reported; 3D: 3 dimensional; AFA: ankle flexion angle; FCA: foot contact angle; SI: strike index; RFS: rear-foot strike; MFS: mid-foot strike; FFS: fore-foot strike; Non-RFS: Non-rear-foot strike; km: kilometre; PB: personal best; *Testing speeds significantly different between RFS and Non-RFS runners: N/A: not applicable; Visual^: foot strike pattern categorized visually based on the following descriptions – rearfoot strike was when the rear part of the foot (or heel) made contact with the ground or treadmill first, midfoot strike was when there was simultaneous contact of the heel and toes together, forefoot strike was when the toes contacted the ground or treadmill first, and non-rearfoot strike was when there was simultaneous landing of the heel and toes together, or if the toes contacted the ground or treadmill first.*

Supplemental Appendix: GRADE Approach

The Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach offers a structured process for developing and presenting summaries of evidence, including its quality, through the use of a transparent framework.[8]

Quality of Evidence Categories

High: Confidence that the true effect lies close to that of the estimate of the effect.[1]

Moderate: Moderate confidence in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.[1]

Low: Confidence in the effect is limited: the true effect may be substantially different from the estimate of the effect.[1]

Very low: Little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of the effect.[1]

Definitions and Justifications for Assessment

The GRADE approach begins by **assessing the study design**, and subsequently assesses various review factors that may alter the quality of evidence.[1] Randomized controlled trials begin with having a "high" rating, while observational studies begin with having a "low" rating.[1] There are five review factors that can potentially rate down the quality of evidence (study limitations, inconsistency, indirectness, imprecision, and publication bias), and three review factors that can rate up the quality of evidence (Large magnitude of effect, dose response, and confounders likely minimize the effect).[1] In the event that a control group is implicit, absent or unclear in a study, this study would begin the GRADE process as having a "very low" rating, as this lack of control would limit the confidence in the effect.[10]

Factors which may rate down the quality of evidence:

Study limitation assessment refers to the assessment of methodological quality through a risk of bias analysis.[7] As described in Supplemental Table 3, studies were assessed under five categories (reporting, external validity, internal validity – bias, internal validity – confounding, and power). Following the GRADE approach guidelines,[7] studies were graded down if, collectively for the outcome; there was a serious risk of bias (-1), or a very serious risk of bias (-2). The scoring criteria for study limitation is outlined below:

Score of 0: No serious risk of bias overall; all studies were considered to have a low risk of bias overall.

Score of -1: Serious risk of bias overall; at least one study was considered to have a moderate risk of bias (and others demonstrated a low risk of bias).

Score of -2: Very serious risk of bias overall; at least one study was considered to have a high risk of bias (and others demonstrated low or moderate risk of bias).

Inconsistency assessment refers to the similarity of effect estimates for each outcome measure across the studies.[2] Study results were considered consistent if the direction, effect size and statistical significance were similar enough to draw the same conclusions.[9] Equally, study results were considered inconsistent if the direction, effect size and statistical significance were dissimilar across the studies, and overall conclusions were difficult to draw. The authors of this review hypothesized that due to the poor quality of

some studies, confidence intervals (CIs) and effect sizes may not be available, and thus the ability to ability to calculate the statistical variation across studies (I^2) would not be feasible. In this case, assessments of inconsistency across different studies would be deemed as not applicable.[10] The scoring criteria for inconsistency is outlined below:

Score of 0: Direction, effect and I^2 all reveal the absence of clinical heterogeneity.

Score of -1: In the case of clinically heterogenous studies, a point was deducted for conflicting or inconsistent results.

Score of N/A: This result was considered if statistical variation across studies (I^2) was unable to be calculated, or if the majority of studies could not provide data to calculate Cis.[10] The lack of information in this domain therefore provides insufficient criteria to make a judgement on consistency across the studies.

Indirectness assessment refers to the examination of differences between study populations, interventions or outcome measures.[3] Two types of indirectness have been reported: one being related to insufficient reporting or descriptions of methods, and the other relating to head-to-head comparisons being unavailable (e.g. no control group).[3] Both types of indirectness are scored independently, but a score of -2 may be given if it is deemed that both types are applicable. The scoring criteria for indirectness is outlined below:

Score of 0: No evidence to suggest indirectness.

Score of -1: Majority of studies for a given outcome measure had no control or comparison group, **OR** the majority of methods described could not be implemented with the same rigour and sophistication as the original study (due to the lack of information provided or the lack of equipment that would be available in a common clinical setting).

Score of -2: Majority of studies for a given outcome measure had no control or comparison group, **AND** the majority of methods described could not be implemented with the same rigour and sophistication as the original study (due to the lack of information provided or the lack of equipment that would be available in a common clinical setting).

Imprecision assessment refers to the examination of CIs to inform the impact of random error on evidence quality.[4] If the CI around the estimate of effect is not sufficiently narrow, the quality of evidence would be rated down. The scoring criteria for imprecision is outlined below:

Score of 0: Sufficient data provided to perform a meta-analysis, ensuring that evidence of precision can be confidently assessed.

Score of -1: Fewer than 75% of the studies presented data that could be included in a meta-analysis,[11] **OR** the majority of studies could not provide sufficient data to calculate CIs and effect sizes.

Publication bias assessment refers to the examination of patterns of results within the published literature.[5] Authors of systematic reviews may suspect publication bias when studies are uniformly small, particularly when studies have been sponsored by the industry.[5] The scoring criteria for publication bias is outlined below:

Score of 0: No evidence of publication bias.

Score of -1: The evidence consists of a number of small studies.

Score of -2: The majority of the small studies are sponsored by industry (or the investigators share another conflict of interest).

Factors which may rate up the quality of evidence:

The GRADE approach suggests the **rating up of quality of evidence** by one level if methodologically rigorous observational studies (which begin the GRADE assessment process as "low") demonstrate high confidence in the estimate of effect.[6] The scoring criteria for rating up the quality of evidence is outlined below:

Score of 0: No evidence of a large or very large magnitude of effect, no evidence that all plausible confounders or other biases could increase our confidence in the estimated effect, and no evidence of a dose-response gradient.

Score of +1: Evidence of a large or very large magnitude of effect, **OR** evidence that all plausible residual confounders or other biases increase our confidence in the estimated effect, **OR** when there is evidence of a dose-response gradient.

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